American Computer Science League

2024-2025 ● Contest 4: Short Problems ● Elementary Division

1. Graph Theory

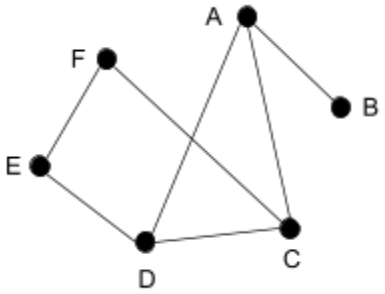
In representing the following friend network as an undirected graph, if Ann's mom is having a party and wants everyone to know at least two other people, which of the following friends shouldn't she invite to the party?

Ann is friends with Beth, Cal, and Dave.
Dave is friends with Ed, Joe, and George.
George is friends with Ann and Ed.
Beth is friends with Hal, Fran, and Ian.
Ian is friends with George and Cal.
Hal is friends with Joe and Ian.

A. Cal
B. Ed
C. Fran
D. Hal
E. Ian

2. Graph Theory

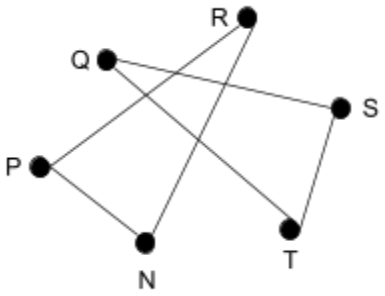
Given the following graph that represents various Interstate Highways between cities, how many different highway routes could you take to get from city B to city C without visiting any city more than once?



A. 2
B. 3
C. 4
D. 6
E. 7

3. Graph Theory

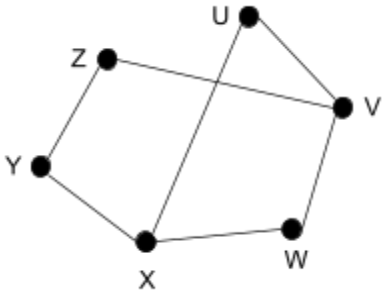
In creating a LAN (local area network), the school's IT director ran fiber optic cable among 3 computers on 2 separate overlapping networks. If she needed to make sure that the computers are all on the same network, how many different single edges could be added to make that possible?



A. 3
B. 6
C. 9
D. 12
E. 18

4. Graph Theory

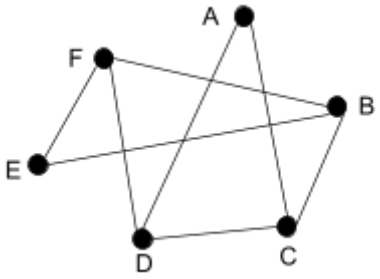
Which of the following is a valid traversal of the graph below?



A. XYZVUYZV
B. XYZVUXYV
C. UVZYXWVU
D. VUXYZVWX
E. VZYXWVZYX

5. Graph Theory

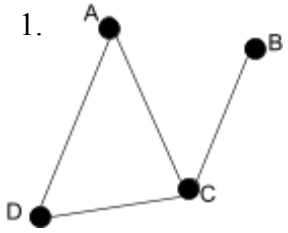
How many cycles are in the following graph?

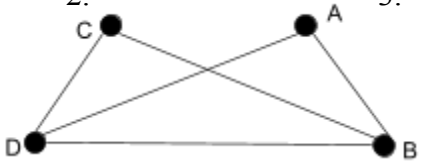


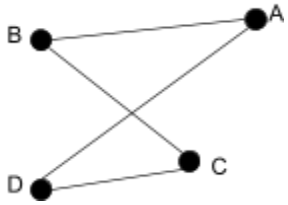
A. 4
B. 6
C. 8
D. 10
E. 12

6. Graph Theory

List all of the following graphs that are traversable:

1. 

2. 

3. 

A. 1, 2
B. 2, 3
C. 1, 3
D. 1, 2, 3
E. Only 1 of them

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1. 图论

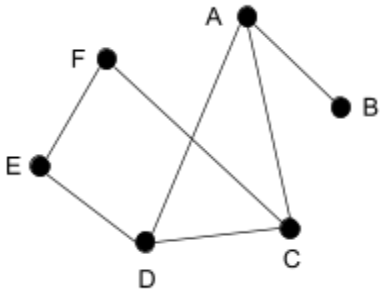
将以下朋友关系网络表示为无向图时，若 Ann 的妈妈要举办聚会并希望每位来宾至少认识其他两人，那么请问她不应该邀请下列哪位朋友？

Ann 是 Beth、Cal 和 Dave 的朋友。
Dave 是 Ed、Joe 和 George 的朋友。
George 是 Ann 和 Ed 的朋友。
Beth 是 Hal、Fran 和 Ian 的朋友。
Ian 是 George 和 Cal 的朋友。
Hal 是 Joe 和 Ian 的朋友。

A. Cal
B. Ed
C. Fran
D. Hal
E. Ian

2. 图论

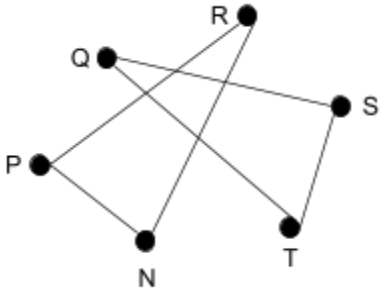
给定如下表示城市间州际公路连接关系的图。请问在不重复经过任何城市的情况下，从城市 B 到城市 C 有多少种不同的公路路线可供选择？



A. 2
B. 3
C. 4
D. 6
E. 7

3. 图论

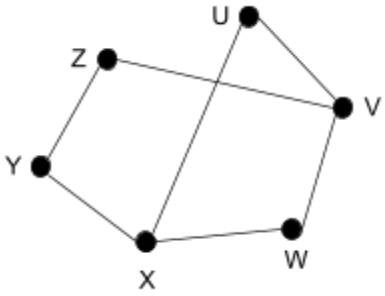
在组建 LAN（**局域网**）**时，**学校的 IT 主任在两组相互重叠的子网中铺设了光缆，每个独立子网连接了 3 台电脑。**若需确保所有**电脑**处于同一网络，**请问**需要添加多少条单边连接才能实现这一目标？**



A. 3
B. 6
C. 9
D. 12
E. 18

4. 图论

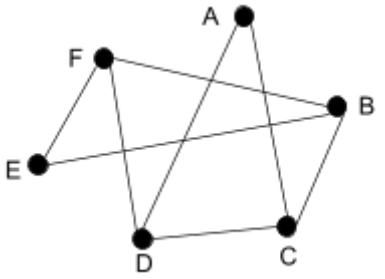
请问以下哪一项是该图的有效遍历？



A. XYZVUYZV
B. XYZVUXYV
C. UVZYXWVU
D. VUXYZVWX
E. VZYXWVZYX

5. 图论

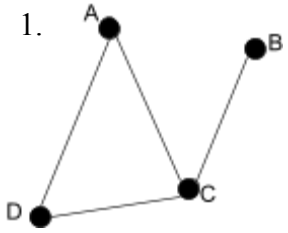
请问下图中有多少个环？



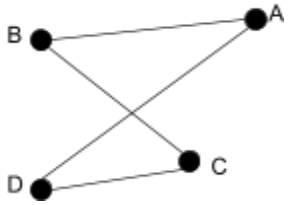
A. 4
B. 6
C. 8
D. 10
E. 12

6. 图论

请列出以下所有可遍历的图：

1. 

2. 

3. 

A. 1, 2
B. 2, 3
C. 1, 3
D. 1, 2, 3
E. 仅有 1 个